



**THE UNIVERSITY
OF QUEENSLAND**
A U S T R A L I A

Fear of temperature

Computational analysis of policy, media
and public climate emotions

Project proposal

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Use of AI Statement

Topic exploration: Yes; AI supported discussion of the topic and candidate approaches.

Research questions: Yes; AI assisted formulation and critique of Section 3.1; I selected the research focus.

Literature review: Yes; AI assisted searches and selected claim/reference checks for Section 2. Reading depth is recorded separately.

Formatting and language: Yes; AI assisted English drafting, translation, editing and document layout.

Other uses: Methodological discussion, planning, Figures 1-9 and export code. No empirical research-data analysis is reported here.

I acknowledge the use of generative AI tools in completing this assessment. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI-generated content, and ensuring it adheres to academic integrity standards.

Table A. AI tool use.

Tool and recorded use	Main activities
ChatGPT; 11 and 15 September 2026	Topic and question development, literature assistance, methodological discussion, writing and language editing.
Gemini; date/version unrecorded	Suggested NLP modules and event/lag analysis; suggestions were assessed and revised by the author.
Codex; 15-16 September 2026	Source and feasibility checks, discussion records, drafting, figures, export code and layout inspection.

Tools: ChatGPT ([OpenAI, n.d.-a](#)), Gemini ([Google, n.d.](#)), Codex ([OpenAI, n.d.-b](#)). Model versions are retained in the working record where available. This declaration concerns proposal preparation; it reports no empirical research-data analysis.

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Figures: 1. Research relationships; 2. Competing mechanisms; 3. RQ evidence routes; 4. Source denominators; 5. NLP pipeline; 6. Structural scales; 7. Measurement and timing; 8. Contextual evidence; 9. Calendar and workload.

Tables: A. AI use; 1. Literature synthesis; 2a-b. Sources and harmonisation; 3a-b. Measures and reference checks; 4. Structural alternatives; 5. Events; 6. Evaluation; 7. Risks; 8. OHS and ethics.

Reading note. This proposal describes planned research; it does not report completed experiments. Numerical examples and diagram trajectories explain the design. Access-dependent choices will be resolved through the documented pilot. The provisional submission date is 16 September 2026; the course deadline is 17 September 2026 at 15:00.

1. Introduction

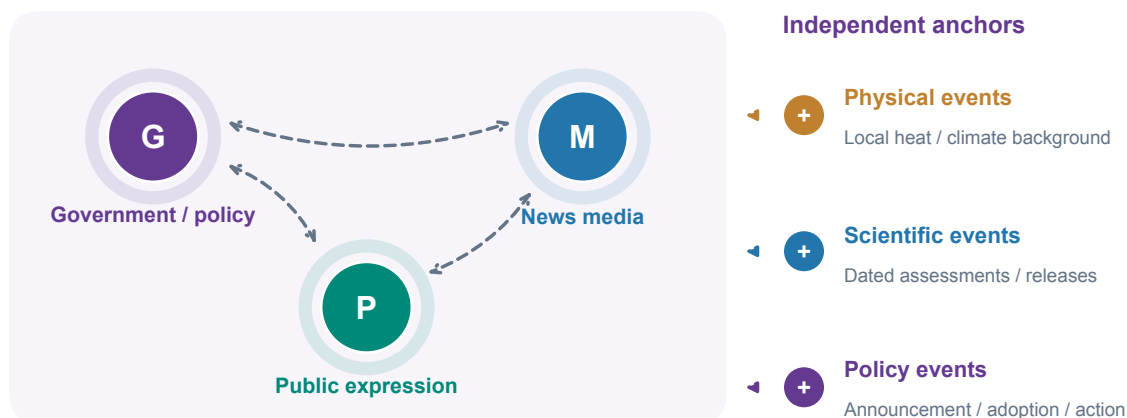
1.1 Motivation and significance

Rising temperature is both a physical process and an object of social anticipation. People may describe immediate danger during a heatwave, worry about the world their children will inherit, or assign responsibility for future disruption. These expressions involve different emotions, time horizons and social relationships. Climate-anxiety research recognises psychological responses to anticipated environmental change as well as direct exposure (Clayton, 2020). Studying how such responses are communicated can illuminate the changing social meaning of warming without assuming that every negative statement expresses fear.

Fear of Temperature asks how warming-related fear and anxiety develop across government and policy discourse, news media and public expression. A central problem is temporal ordering: does public concern follow physical events, institutional communication or media attention, and when might it precede them? Earlier work links climate concern to political cues, media and other conditions (Brulle et al., 2012). Research on affective agenda dynamics also demonstrates the relevance of comparing government, media and public expressions (Zhou et al., 2023). Figure 1 frames these as competing relationships to investigate rather than a predetermined chain of influence.

One social question, three discourse roles

How do warming-fear expressions change across roles and time?



Dashed arrows: competing relationships, not established effects.

Figure 1. Three discourse roles. Dashed links are candidate relationships, not causal findings.

1.2 A methodological transition

This study investigates warming-fear expressions through validated semantic similarity and structural analysis. Dictionary and n-gram exploration develops into traceable collection, pretrained embeddings and comparisons across roles/time. The thesis remains the main outcome.

Verified reference passages retain the focus on fear, including anticipated harm to families and future generations. Similarity retrieves candidates; contextual checks validate interpretation. Separate emotion classification and semantic-role extraction are outside scope. Lexical baselines test whether pretrained representations improve retrieval.

1.3 Scope and historical framing

The core collection will prioritise English-language texts associated with the United States, European settings, Australia and New Zealand. Broad geographical coverage is an objective, while the main comparison concerns source roles and identifiable communities.

Chinese-language material, East Asian settings and AOSIS members are extension candidates rather than dependencies of the core study. Publication location, author location and the place discussed in a text will be distinguished. English language alone will not establish geographic or cultural identity.

The target collection begins in January 1988 and extends to a documented cutoff in 2026. The establishment of the IPCC in 1988 motivates an institutional anchor ([Intergovernmental Panel on Climate Change, n.d.](#)). It does not establish an emotional turning point. Recoverable material from 1938 onwards may form an earlier contextual layer, with Callendar's work providing a scientific-historical reference ([Callendar, 1938](#)). A claim about change across 1988 requires adequate observations on both sides; the main collection alone cannot demonstrate that transition.

Historical span and analytical coverage are therefore different. Each comparison will use a sufficiently dense common window, with incomplete months, archive gaps and changes of source explicitly recorded. Letters to editors and early forums may extend the record of public expression, but their selection processes differ from contemporary platforms. The intended inference concerns the sampled discourse. Claims about population emotion require additional sampling or validation evidence.

Longer-term anticipated consequences are the substantive priority. Direct heat danger will be collected alongside them. Reference passages and validation notes distinguish threat, time horizon and affected group without requiring automatic corpus-wide emotion labels. Fear can concern the future, and collective anxiety can arise during a present emergency. Cultural heritage may inform discussion, but the project does not presuppose heritage status.

1.4 Broad aim and intended contribution

The broad aim is to understand how expressions of warming-related fear and anticipated harm are organised across policy, news and public discourse, and how their similarity and temporal relationships vary around physical and institutional events.

Technical resource. A versioned, traceable corpus and reproducible pipeline with documented coverage, denominators, sampling weights and a data dictionary. Code, configurations and random seeds will support reproduction; only permitted material will be released.

Validated method. An evaluated workflow connecting reference-guided semantic retrieval, similarity structure and temporal comparison, with lexical baselines and source/period checks. Any algorithmic novelty claim depends on demonstrated improvement, not simply combining existing models.

Empirical contribution. A thesis investigating the timing, convergence and organisation of warming-fear expressions through statistical estimates and contextual evidence, with explicit causal limits and potential for publication. Small or uncertain associations remain legitimate findings; success does not require a dominant driver or significant policy effect.

2. Background and Literature Review

2.1 Climate emotions and anticipated futures

Climate anxiety concerns responses to possible as well as experienced environmental harm. Clayton cautions against assuming that concern is necessarily maladaptive (Clayton, 2020). Solastalgia instead describes distress associated with environmental change in a place to which people remain attached (Albrecht et al., 2007). These accounts connect emotion to anticipated futures and lived environments, but neither makes all negative climate language equivalent to anxiety.

Pihkala's taxonomy distinguishes fear, worry, grief, anger, guilt and hope (Pihkala, 2022). For this project, that diversity argues against a single scale running from bodily panic to existential anxiety. Reference selection and validation must distinguish emotional expression, target, time horizon and affected group. Textual worry does not establish a persistent psychological condition; keyword cues require contextual validation.

O'Neill and Nicholson-Cole distinguish fear-inducing climate imagery from effective engagement (O'Neill & Nicholson-Cole, 2009). Threatening representations need not indicate policy support. Validation must separate issue relevance, fear expression and agreement; similarity alone cannot resolve them. Textual expressions are not direct observations of population mental health.

2.2 Agenda-setting and responsibility frames

Agenda-setting addresses relationships between the prominence of issues in media and public agendas (McCombs & Shaw, 1972). A corpus share, however, measures the prominence of an issue in sampled output; without audience information it does not measure exposure or establish whose agenda changed.

Framing concerns how communication selects a problem, causal interpretation, evaluation and response (Entman, 1993). Increased climate coverage can frame warming as economic disruption, physical danger or intergenerational injustice. These are different interpretations even if their overall issue attention is similar. Cause, blame and response duty also differ: an actor may be asked to respond without being blamed for creating the problem. Contextual reading is therefore needed to interpret proximity: the project will not infer cause, blame or duty automatically from a similarity edge.

Similar policy and public language may reflect adoption, quotation or opposition; convergence alone does not establish agreement.

Table 1. Literature and design.

Evidence strand	What it contributes	Requirement for this study
Climate emotions	Multiple responses to threat	Verify fear, target and horizon
Agenda and framing	Prominence versus interpretation	Measure attention and narrative content
Climate concern	Elite/media and physical explanations	Compare roles and exposure scales
Contextual NLP	Retrieval and text representations	Validate by source and period
Temporal methods	Ordering and event-related changes	State estimands and causal limits

2.3 Evidence from climate concern and role comparisons

Brulle and colleagues construct quarterly U.S. climate-concern measures from 74 surveys during 2002–2010, comparing weather, science information, media and political factors (Brulle et al., 2012). Their findings motivate institutional explanations, but aggregate concern is not specifically future-oriented fear. National quarterly observations may also conceal responses to local, short-lived heat exposure. The proposed text measures add narrative detail and shorter eligible windows; they introduce selection and representation errors that survey-based concern does not share in the same form. They should complement this evidence rather than be assumed more accurate.

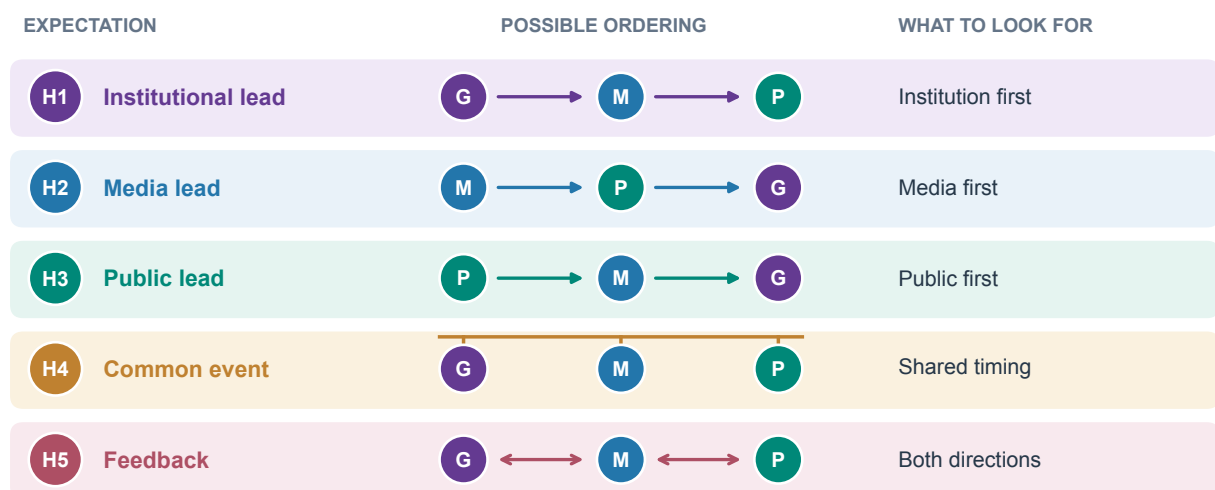
Zhou and colleagues compare public, government and media emotional communication on Weibo during COVID-19 (Zhou et al., 2023). This is a close precedent for role-based temporal comparison, but one platform and a pandemic period differ from a historical climate corpus spanning institutions and media systems. The present design retains the role comparison while testing source continuity, emotional targets and independent physical observations. Neither precedent establishes that a fixed direction of influence applies across all periods.

2.4 Competing temporal mechanisms and expectations

Institutional signals can supply authoritative risk interpretations (Brulle et al., 2012); media can select and frame issues (Entman, 1993; McCombs & Shaw, 1972); public expression may supply news material and pressure for response. Figure 2 makes the competing temporal expectations visible. H3 is a project hypothesis informed by role-comparison research (Zhou et al., 2023), not an established climate result.

Five competing temporal patterns

Read each row as a candidate explanation; routes may differ across windows.



G: government/policy M: media P: public · Predictive ordering alone is not causation.

Figure 2. Competing temporal mechanisms. Arrows indicate candidate orderings; colours identify roles.

The same ordering can arise from anticipation, shared releases, reposting or changing participants. A coarse monthly bin can hide a faster ordering. Local heat exposure also differs from global climate background. These expectations may coexist across windows; temporal prediction alone cannot identify social feedback or individual emotional contagion.

2.5 Embeddings, semantic retrieval and structural analysis

Sentence-BERT supports efficient sentence comparison (Reimers & Gurevych, 2019). The proposed primary encoder, all-MiniLM-L6-v2, maps sentences and short paragraphs to 384-dimensional vectors and is trained for sentence-level similarity. Its default 256-word-piece input limit makes passage segmentation essential (Sentence Transformers, n.d.). Similarity may recover paraphrases missed by exact words, but it is neither emotional intensity nor agreement. Lexical/TF-IDF retrieval will provide a transparent baseline.

ClimateBERT is a bounded domain-representation comparison, not a second mandatory model-development project. The candidate distilroberta-base-climate-f is a climate-adapted masked-language model; it is not automatically a sentence-similarity model (ClimateBERT, n.d.). Any comparison will specify the checkpoint, pooling and input treatment, then assess retrieval on the same judged passages. Its vectors will remain in a separate index; scores from different encoders will not be treated as directly interchangeable.

FAISS provides vector indexing and nearest-neighbour search (Faiss, n.d.). It neither interprets fear nor builds the substantive relationship graph. A similarity-derived tree or graph represents proximity under chosen construction rules. Its branches are not observed pathways of influence. Actual reply or citation links, if available, are separate edge types rather than substitutes for semantic similarity.

A limited reference set and explicit counterexamples connect representation to the social question. Contextual checking is retained, while independent emotion classifiers, SRL, automatic responsibility extraction and BERTopic are outside the core implementation. The main technical work is corpus harmonisation, retrieval evaluation and reliable structural comparison rather than training a new language model.

2.6 Selected research gap

The selected gap is the connection between changing discourse prominence and the organisation of warming-fear expressions across policy, media and public sources. Counts alone cannot show whether these roles use similar expressions of anticipated harm; proximity alone cannot establish that they express fear. This study connects verified fear/worry reference passages to evaluated retrieval, then compares similarity patterns and their timing within common coverage windows. Representative and contradictory passages support interpretation.

Temporal models address bounded questions. Granger analysis concerns added prediction (Shojaie & Fox, 2022); ITS estimates level and slope changes (Lopez Bernal et al., 2017), while a temporal cutoff alone does not identify a causal effect (Hausman & Rapson, 2018). Stable source differences also complicate cross-lagged interpretations (Hamaker et al., 2015). The contribution is a reproducible empirical analysis of expression and structure, not proof of population anxiety or an assumed direction of influence.

3. Project Plan

3.1 Research questions

RQ1 - Who changes first? How do policy, media and public warming-fear expressions converge or diverge over time, and which role changes first? Compare validated reference-similarity series using CCF and, where supported, conditional VAR with an own-history predictive baseline.

RQ2 - What changes around events? How do similarity distributions and supported structural summaries change around independently dated physical, scientific and policy events? Segmented regression/ITS tests prespecified scalar outcomes, with window sensitivity and placebo-date checks.

RQ3 - How are fear expressions organised? What groups and connections characterise warming-related fear and anticipated harm, and how are they reorganised across roles and time? A documented similarity structure and original passages support interpretation; stability checks test dependence on references and construction choices.

The focus remains anticipated warming harm. Similarity measures expression proximity, not psychological intensity.

Research questions → computational tasks → evaluation

Three questions share one corpus, representation and validation protocol.

QUESTION	COMPUTATIONAL TASK	ANALYSIS + EVALUATION
RQ1 Who changes first?	Reference similarity → S / Q by source role	CCF / conditional VAR → Lags + held-out predictive gain
RQ2 What changes at events?	Similarity + structure → S / Q; fixed summaries	Segmented regression / ITS → Level / slope + window checks
RQ3 How is fear organised?	Similarity structure → Groups + source passages	Stability + context → Reference / source / period checks

Shared safeguards: fixed units + weights · source/time validation · original context

Figure 3. Research questions and evidence routes. Structural choices and temporal outcomes are fixed after the pilot.

3.2 Evidence-driven implementation

The bounded pilot tests collection, dates/roles, references, retrieval and one structural representation. Choose a common window by coverage; measured processing and review costs govern expansion.

Freeze references, encoder, units, weights, construction and outcomes before comparison. Retain baselines and counterexamples. ClimateBERT and wider coverage are extensions.

3.3 Selected sources and acquisition sequence

The initial sources are GOV.UK, Guardian and the preferred Reddit application route. Table 2a fixes priorities, not access approval; wider geography follows a validated pilot.

Table 2a. Selected sources and access gates.

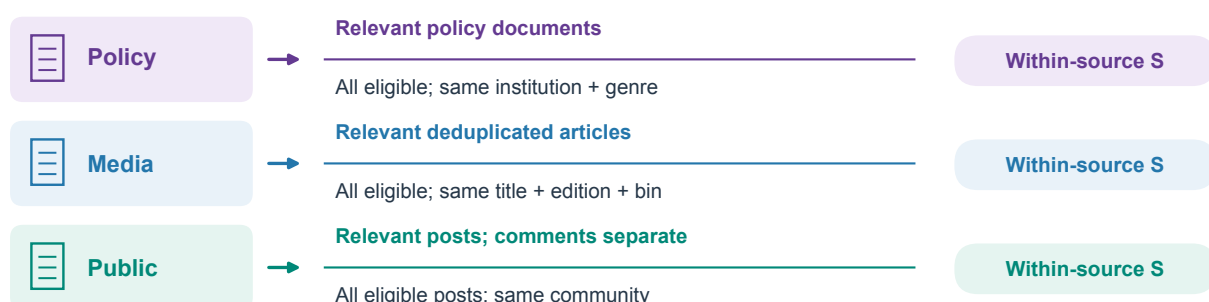
Source / priority	Role and route	Coverage / denominator	Current gate
GOV.UK / primary	Policy; Search + Content APIs	Institution/type/month; indexed documents	Metadata pilot verified; body pending
Guardian / primary	News; Open Platform API	Accessible dated articles; same filters without topic query	Key, rights and extraction check
Reddit / preferred	Public; Researchers programme	Approved community/window; posts and comments separate	Application; permitted processing route
Mastodon / fallback	Public; permitted instance/API	Verified window; no historic completeness assumed	Instance rights, access and counts
UNFCCC / IPCC	Policy / science context	Dated releases; not all-policy denominators	Item-level rights check
US/EU/AU/NZ archives; licensed news; letters/forums	Extension strata	Source-specific accessible windows	Added only after core validation

GOV.UK has a public Search API ([Government Digital Service, n.d.](#)). M1.1 verified nine July 2026 DEFRA policy-paper metadata records: a bounded document denominator, not verified historical coverage or body-use permission.

Guardian offers a dissertation developer route with article text and a 500-call/day limit ([Guardian News & Media, n.d.](#)); verify terms and all-topic counts. Reddit approval must cover processing/export and reviewer access ([Reddit, n.d.](#)). If unavailable, assess one permitted Mastodon community ([Mastodon, n.d.](#)).

Three source roles, three auditable denominators

General attention comes from the eligible collection, not climate-only retrieval.



Climate-only subsets support similarity analysis, not general attention estimates.

Figure 4. Source denominators. Each ratio uses the same eligible collection and unit; posts and comments remain separate.

Freeze source selection before inspecting fear scores. The 1988-2026 collection target does not imply continuous analytical coverage.

3.3 Corpus normalisation and historical comparability

The principal risk is analytical harmonisation across policy, media and public material, not merely database normal forms. SQL or NoSQL may preserve heterogeneous raw records; all analytical batches must satisfy the same documented field, unit and provenance contract. Vector normalisation cannot make publication systems comparable.

Table 2b. Cross-role and historical standards.

Dimension	Prespecified rule	Failure response
Three source roles	Assign by publishing function; retain genre and scientific subtype	Mixed/unknown stays separate
Voice and attribution	Store quoted speaker independently of publishing role	Do not move quoted speech between series
Dates and versions	Preserve first publication, update and collection dates; use first publication for bins	Unknown date excluded from timed analysis
Units and weights	Documents for source counts; passages for embeddings; retain parent IDs	Average within document before equal-document Q
Duplicates / joint issues	Count each document once in pooled totals; retain all institution links	Fractional allocation when combining institution strata
Source mix	Freeze source/genre weights in common windows; report coverage	Compare fixed-source subsets; no silent backfill
Channel transitions	Register archive/platform/format breaks independently of scores	Separate periods unless overlap supports comparison

Government means institutional policy output, media means editorial publication, and public means non-institutional expression in eligible community channels. Official accounts on social platforms retain an institutional role; news accounts retain media identity. Uncertain authorship remains unknown. IPCC assessment texts have a scientific subtype and cannot define all-government salience.

S uses native document/post units with matching denominators. For Q, score relevant passages, average within each document/post, then apply source sampling weights; this prevents long reports dominating solely through passage count. In Equations (1) and (3), i denotes a parent unit and the bar in Equation (3) its mean relevant-passage proximity; p in Equation (2) indexes passages. Other aggregation choices are declared sensitivity analyses.

Older newspapers, letters, forums and contemporary platforms will not be concatenated as interchangeable populations. A source-era register will record archive scope, editorial selection, article type, digitisation/OCR, platform rules and observed coverage. Select common monthly bins or shared non-overlapping quarters; missing coverage is not zero expression. Publication location is not an author's location.

The acceptance gate requires auditable role/date/unit mappings, deduplication and denominators for every analysed stratum. Historical volume below 100 flags review, not automatic exclusion: effective counts and uncertainty govern usable resolution. Use the channel-transition diagnostic in Section 3.6 and fixed-source comparisons; failed comparability narrows the window rather than being repaired by a numerical rescaling.

3.4 Operational definitions and construct validity

The object is expressed fear or worry about warming, with anticipated harm as the priority. Verified reference passages will cover immediate threat, family/intergenerational concern and wider social futures. These are overlapping retrieval perspectives, not a clinical scale or mutually exclusive automatic labels. Similarity alone cannot distinguish a concern from a quotation or its rejection.

Table 3a. Constructs and measures.

Construct	Operational measure	Interpretation boundary
Attention S	Weighted warming-relevant share	Requires an eligible background denominator
Proximity q	Mean cosine to fixed fear references	Semantic proximity, not fear intensity
Temporal Q	Weighted mean q in relevant units	Conditional on source and retrieval coverage
Structure	Prespecified tree/graph summary	Constructed proximity, not diffusion
Context	Reviewed representative and contrary passages	Interpretation, not automatic attribution

In Equation (1), U contains eligible parent documents/posts for source r and time t ; w denotes sampling weights and R a validated warming-relevance decision. A parent is relevant when at least one reviewed or validated-rule passage qualifies; this rule is fixed within comparisons. Without an issue-independent collection or known sampling weights, S is not estimated.

$$S_{rt} = \frac{\sum_{i \in U_{rt}} w_i R_i}{\sum_{i \in U_{rt}} w_i} \quad (1)$$

For the passage-level specification in Equation (2), z_p is a passage embedding and A the fixed reviewed fear/worry reference set. The equation averages cosine similarity to that set. References are kept balanced across the intended concern perspectives; prespecified subsets can be inspected separately without choosing them to maximise a temporal effect.

$$q_p = \frac{1}{|A|} \sum_{a \in A} \cos(z_p, z_a) \quad (2)$$

$$Q_{rt} = \frac{\sum_{i \in U_{rt}} w_i R_i \bar{q}_i}{\sum_{i \in U_{rt}} w_i R_i} \quad (3)$$

In Equation (3), the bar denotes the mean of relevant passage scores within parent i . Q is therefore document-weighted reference proximity, not emotion probability or prevalence. Empty denominators are missing. If only retrieved candidates are retained, report their conditional distribution rather than an all-text estimate. Fixed units and weights support within-series comparisons; different publication genres cannot be ranked as more fearful from their raw score levels.

3.4 Reference design and validation (continued)

Reference passages must explicitly connect concern to warming and retain enough context to identify target, horizon and speaker. Real permitted passages may be supplemented by clearly labelled synthetic development examples. References and their source documents must not leak into the held-out evaluation set; duplicate clusters remain together. Document who selected each reference and why.

Table 3b. Reference and counterexample checks (synthetic).

Synthetic passage	Intended check	Interpretation boundary
I worry about my children's future.	Future concern	Climate context is missing
I fear warming will harm my children.	Warming-related anticipated harm	Candidate positive reference
The government said people should not panic.	Quotation and negated advice	Not government fear
Scientists warn heatwaves are deadly.	Risk-only counterexample	Danger is not expressed fear
If warming worsens, I fear our town will become unsafe.	Conditional anticipated harm	Present concern; future condition
I am not afraid of warming; I oppose this tax.	Negation and policy disagreement	Similar vocabulary can mislead

The validation task checks whether retrieved passages support the intended interpretation; it does not require training a separate classifier. Review top-ranked results, difficult counterexamples and a background sample to expose missed expressions. Distinguish publishing role from quoted speaker and emotion-holder. Quoted public concern remains in its publication source series.

Vary reference wording and membership on development data to detect a result driven by one phrase. Freeze the selected set for final comparisons and report sensitivity transparently. A high cosine score is not a fear label: if contextual checks fail, restrict claims to topical proximity and revise the references before proceeding.

For structural interpretation, inspect representative and contradictory passages from each reported group. Cause, blame or duty may be discussed when explicit in those passages, but will not be automatically extracted or inferred from the shape of a tree.

3.5 Representation, retrieval and structure

Use all-MiniLM-L6-v2 as the primary sentence encoder ([Sentence Transformers, n.d.](#)), with lexical/TF-IDF retrieval as a baseline. Segment long texts before encoding and preserve parent-document context. Freeze model revision, pooling, passage rules and reference vectors within each comparison. Unit-length vector normalisation supports cosine computation; it does not harmonise dates, source populations or sampling denominators.

Fear references → embeddings → retrieval → structure

One bounded workflow; original passages remain linked to every result.

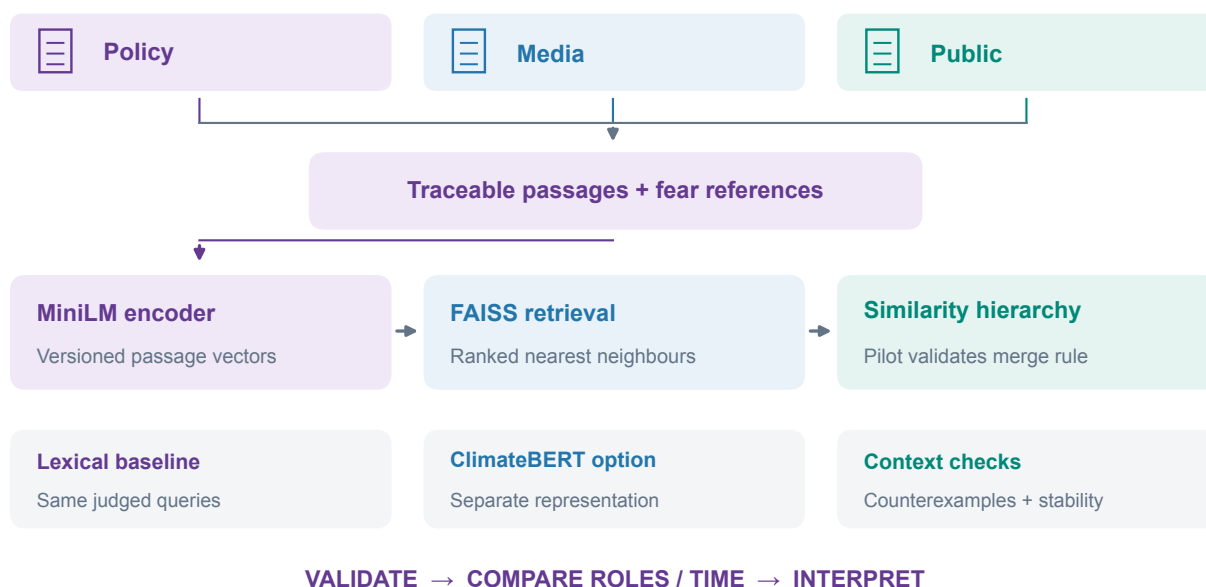


Figure 5. Similarity workflow. MiniLM produces vectors; FAISS retrieves neighbours; validation supports structural and temporal interpretation.

Retrieval. Encode queries/references with the same encoder as corpus passages. FAISS indexes vectors and returns neighbours ([Faiss, n.d.](#)). Start with exact search on the bounded pilot; approximate indexing is a scale option whose neighbour recovery must be checked against exact results. Record throughput, memory and runtime. ClimateBERT is a separate, optional representation comparison with documented pooling and the same evaluation material ([ClimateBERT, n.d.](#)).

Structural representation. The initial pilot will use an average-linkage hierarchy on cosine dissimilarity for a capped, source-balanced passage sample ([SciPy Developers, n.d.](#)). Leaves retain passage IDs; internal nodes represent merges, not causes. A sparse nearest-neighbour graph is the scale fallback if the full pairwise distance calculation exceeds resources. Section 3.5.1 compares alternatives and sets their boundaries. Observed replies/citations remain separately typed; FAISS index-internal connections are not discourse relationships.

Comparability. Use matched source coverage and sampling; preserve duplicate clusters. Review fear-reference enrichment and role mixing within groups, with source-balanced resampling. Freeze the cut rule and any scalar summary used for RQ2 before final tests. Reference-set and construction sensitivity distinguish robust structure from an arbitrary partition.

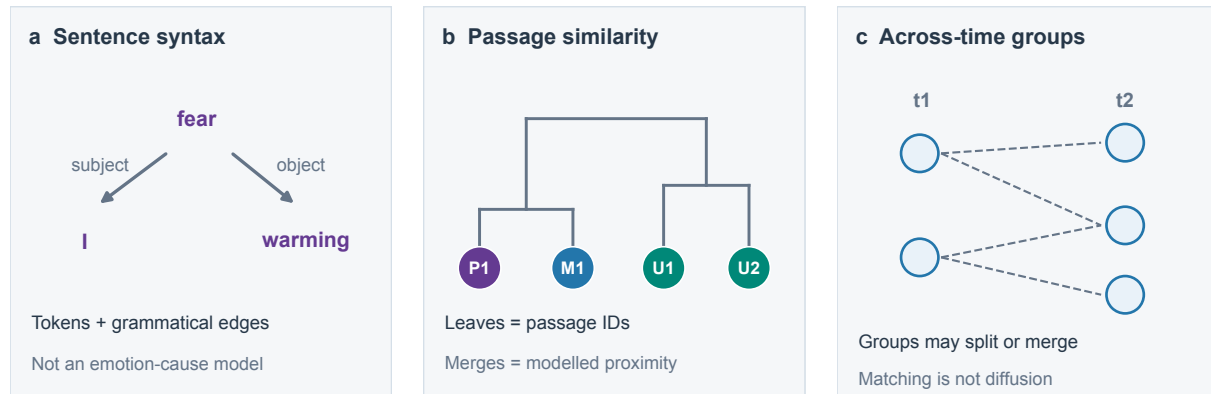
No separate emotion classifier, SRL system or topic model is a core deliverable. Human review is bounded validation and contextual interpretation, not exhaustive manual coding.

3.5.1 Tree and graph designs: scope and meaning

Different trees encode different relations. Figure 6 contrasts sentence structure, passage proximity and temporal group matching. These are design sketches, not observed findings; a branch never establishes psychological causation.

Different structures encode different relations

Design sketches only: grammar, similarity and temporal matching are distinct.



Core: similarity hierarchy. Syntax is diagnostic; temporal matching is conditional.

P / M / U: policy / media / public. All examples and connections are synthetic.

Figure 6. Three structural scales. Synthetic nodes illustrate syntax, similarity and time-linked groups; line meanings differ.

Table 4. Structural alternatives and scope.

Method	Nodes / relation and use	Commitment / limitation
Similarity hierarchy	Passages; average-linkage cosine proximity	Core pilot; quadratic distance cost
Dependency tree	Tokens; grammatical dependencies	Optional difficult-case check; not emotion attribution
Constituency tree	Phrases; syntactic containment	Optional scope check; parser errors remain
WordNet / ontology	Word senses; lexical/concept relations	Vocabulary aid; not an eco-anxiety taxonomy
Causal event tree	Event, harm, expression; claimed narrative links	Outside core; needs separate evidence/annotation
Topic/group evolution	Groups across time; matched representations	Conditional extension; splits/merges form a graph

Dependency and constituency parsers describe grammatical relations and phrase structure (Stanford NLP Group, n.d.-a, n.d.-b). A small diagnostic sample can check attachment, negation and quotations; syntax alone does not recover who feels fear or why. WordNet links senses through lexical relations (Princeton University, n.d.); it does not justify placing despair and worry under a clinical eco-anxiety parent.

Time-linked groups use a fixed encoder and matching rule; retain unmatched groups. Splits/merges are modelled correspondence, not diffusion. Causal trees require explicit narrative evidence and cannot reveal psychological mechanisms. The similarity hierarchy is core; other methods require demonstrated need and spare capacity.

3.6 Temporal alignment and analysis

Use common coverage and frequency: monthly, daily/weekly for dense heat pilots, or quarterly for sparse institutions. Recompute ratios from counts; never duplicate quarters into months. GISTEMP describes regional climate; station/ERA5 data address exposure ([European Centre for Medium-Range Weather Forecasts, n.d.](#); [NASA Goddard Institute for Space Studies, n.d.](#)). Record location evidence, baselines and versions; publication location is not exposure.

RQ1: ordering and conditional prediction. Address trends, seasonality and serial dependence before CCF analysis. Equation (4) correlates processed series X and Y; positive k means X precedes Y. Compare role-specific S or Q under a fixed reference set and compatible coverage.

$$C_{XY}(k) = \text{Corr}(X_t, Y_{t+k}) \quad (4)$$

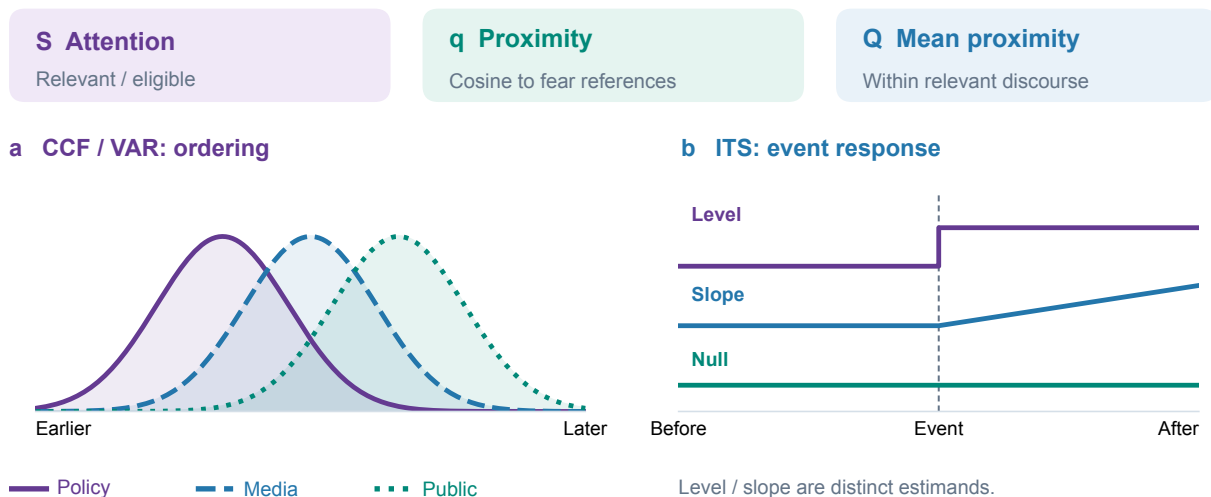
Where coverage and diagnostics permit, a small vector autoregression (VAR) tests added prediction beyond the outcome's own history ([Shojaie & Fox, 2022](#)). Report direction, lag range, uncertainty and held-out gain. CCF peaks alone do not select lag order. Use a small prespecified set of role-specific S or Q series; avoid redundant measures.

RQ2: event-associated change. Segmented regression/interrupted time series (ITS) estimates level and slope changes ([Lopez Bernal et al., 2017](#)). In Equation (5), t is centred on onset and D switches from 0 to 1; β_2 and β_3 estimate level and slope changes. Diagnose serial dependence in u and adjust uncertainty.

$$Y_t = \beta_0 + \beta_1 t + \beta_2 D_t + \beta_3 (tD_t) + u_t \quad (5)$$

Measure the discourse, then test its timing

Illustrative trajectories only; no observed data or estimated effects.



Inputs: aligned S / Q and event dates. Outputs: lags, predictive gain and level / slope.

Figure 7. Temporal analysis. All curves are schematic, not observed results.

Event selection. Select independently dated events before inspecting outcomes; require common coverage and auditable denominators. Pilot one event, distinguishing announcement, adoption, implementation and scientific release.

3.6 Event selection and narrative interpretation (continued)

Table 5. Candidate events.

Candidate / type	External date	Window / outcome	Comparison and limit
Kyoto adoption / policy	11 Dec 1997	Initially ± 6 months; S/Q level and slope	Historical overlap unverified
Paris adoption / policy	12 Dec 2015	Initially ± 6 months; S/Q level and slope	Anticipation; concurrent coverage
Heat episode / physical	○ Independent onset, peak and end	Dense local window; reference proximity	Matched exposure; seasonality
IPCC release / scientific	○ Report-specific release date	Common coverage; S/Q or fixed structure summary	Shared shock; media spillover

Adoption dates follow UNFCCC records (United Nations Framework Convention on Climate Change, n.d.-a, n.d.-b). Extend initial six-month windows if trends, seasonality or serial dependence require longer baselines. Fix windows and aggregation before final tests using coverage and baseline adequacy.

Controls require comparable pre-event measurement and defensibly different exposure. Register anticipation/concurrent shocks; inseparable events share a window. Placebo dates and alternative windows assess sensitivity, not causal identification (Hausman & Rapson, 2018).

Channel-transition diagnostic. Apply Equation (5) to independently documented corpus/channel transitions (e.g., letters coverage ending, forum uptake, Reddit eligibility), using comparable observations on both sides. Check S/Q level and slope changes against fixed-source subsets. Coincident jumps flag possible composition bias, not proof of artificial discourse change. Without comparable overlap, document the coverage break and analyse strata separately.

RQ3: organisation and interpretation. Examine the selected similarity structure and reference-linked expression groups under matched coverage. Read representative and contradictory passages to interpret threatened futures and contextual meanings. Shared wording may reflect quotation or duplication; neither a branch nor a time-ordered similarity edge establishes adoption, influence or responsibility.

Fear remains the reference, not an assumed model label

Synthetic examples; topical similarity alone cannot establish fear.

I fear warming will harm my children.

Candidate reference: anticipated harm

Heatwaves are deadly.

Counterexample: risk without expressed fear

I am not afraid of warming.

Counterexample: explicit negation

Inspect context → validate retrieval → interpret groups with original passages.

Figure 8. Reference validation. Synthetic examples distinguish fear expression from thematic similarity and negation.

3.7 Evaluation and validity

A role-balanced development batch of approximately 150 passages will estimate ambiguity and review costs, not final performance. Define relevance to the fear/worry reference perspectives, including risk-only and negated counterexamples. Size a separate source/period-stratified test set from the pilot budget and desired precision. Record enrichment and assess background misses.

Recruit an independent reader unfamiliar with the project, who did not select references or develop models. Provide a neutral task guide, but conceal preferred hypotheses, model identity and the author's initial judgements where feasible. The supervisor is not the sole validation reader. Record agreement and disagreements before discussion. If unavailable, report single-reader checking and restrict claims. Split by document/duplicate cluster; exclude reference-source material.

Table 6. Evaluation plan.

Task	Baseline / reference	Comparison	Metric / output	Validation setting
Retrieval	Lexical / TF-IDF	MiniLM; optional ClimateBERT	Precision@k; nDCG@k	Held-out query/passage judgements
Construct link	Contextual judgement	References and counterexamples	False matches; missed expressions	Source/period audit; background sample
Structure	Fixed construction	Limited parameter/resampling checks	Neighbour/group stability	Matched coverage and sample size
RQ1	Own-history model	Conditional VAR + roles	Held-out MSE gain; lag uncertainty	Common window; chronological holdout
RQ2	Pre-event trend	Segmented regression / ITS	Level/slope; uncertainty	Fixed outcomes; placebo dates
RQ3	Original passages	Structural groups and contrasts	Supported interpretations	Representative and contrary evidence

Ranked evaluation needs documented relevance judgements. Recall will only be reported where sufficiently complete relevant-item judgements support its denominator; a reviewed top-k list alone does not establish full-corpus recall. Retrieval scores do not validate fear intensity. Any approximate FAISS search is separately checked for neighbour recovery against exact search.

Use reference-set sensitivity, resampling and source/period checks to assess whether conclusions survive representation and construction choices. Unreliable fear-related distinctions trigger revised references or narrower claims, not automatic replacement with an unvalidated cosine score. Account for source, time, author and duplicate dependence.

Freeze outcomes, thresholds, weights and lag ranges before final comparisons; alternatives remain sensitivity/exploratory analyses. Non-significance proves neither absent physical influence nor exclusive policy control (Wasserstein & Lazar, 2016).

3.8 Project risks and scope adjustment

Cross-role normalisation and historical comparability are the highest-priority risk. Access, model suitability and interdependent implementation can also delay analysis. Ratings below express management priority; a new model release is low risk because replacing a validated encoder is optional.

3.8.1 Risk assessment

Table 7. Risks, triggers and fallbacks.

Risk	Trigger / consequence	Rating	Concrete response
R1. Normalisation / history	Incompatible roles, units or source eras	High	Apply Table 2b; fixed-source windows; separate incompatible strata. Freeze core corpus by 18 Dec.
R2. Model / tuning cost	Fear counterexamples fail; pilot budget exceeded	High	Retain lexical baseline; cap representation trials. Fine-tuning is optional and must fit a written time/compute budget.
R3. Better model release	New checkpoint appears after freeze	Low	Log it; retain frozen encoder. Compare only at a planned gate; never mix vector spaces.
R4. Practical conflicts	Schema, segmentation or encoder changes break downstream outputs	High	Version interfaces and dependencies; integration checks; rebuild all affected descendants.
R5. Human-data ethics	Approval, access or release conditions unresolved	High	Pause affected data use; seek UQ determination; continue permitted metadata/synthetic work.
R6. Unknown measurement issue	Invalid scores, unstable structures or time diagnostics	High	Roll back to last validated checkpoint; log cause; revalidate changes using fresh held-out evidence.
R7. Access / sparse coverage	Preferred source denied or common bins inadequate	High	Test permitted Mastodon fallback; shorten window or use shared quarters; no population-level splicing.
R8. Compute / review	Pairwise costs or independent review exceed capacity	Medium	Cap balanced sample; sparse graph fallback; narrow claims if independent review unavailable.
R9. Disruption / loss	Illness or corrupted artefacts	Medium	Tested backups and protected contingency; prioritise validated core work.

3.8.2 Dependencies, rollback and review gates

Stages overlap: raw records → units → vectors/index → scores/structure → estimates. Upstream changes invalidate dependent outputs. A manifest links data, code, model and references. Preserve failed runs and validated snapshots; never mix incompatible versions.

Rollback responds to invalid measurement or implementation, not inconvenient or non-significant findings. Post-outcome redesign is marked exploratory and needs fresh held-out validation. Gates: 16 Oct pilot; 18 Dec core corpus; 15 Jan measurement freeze; 28 Feb all analyses. Missing a gate triggers reduced coverage or optional methods, preserving fear validity and the 10-20 h/week budget.

3.9 Milestones, resources and time allocation

At 10-20 hours/week, the provisional budget is 450-590 hours plus 15% contingency. All analysis and robustness checks must finish by 28 February 2027. The 320-400 hours allocated before March require about 14-17 hours/week before contingency; sustained lower capacity will trigger early scope reduction.

A research calendar with protected analysis time

10–20 hours/week · planned task effort 450–590 h + 15% contingency

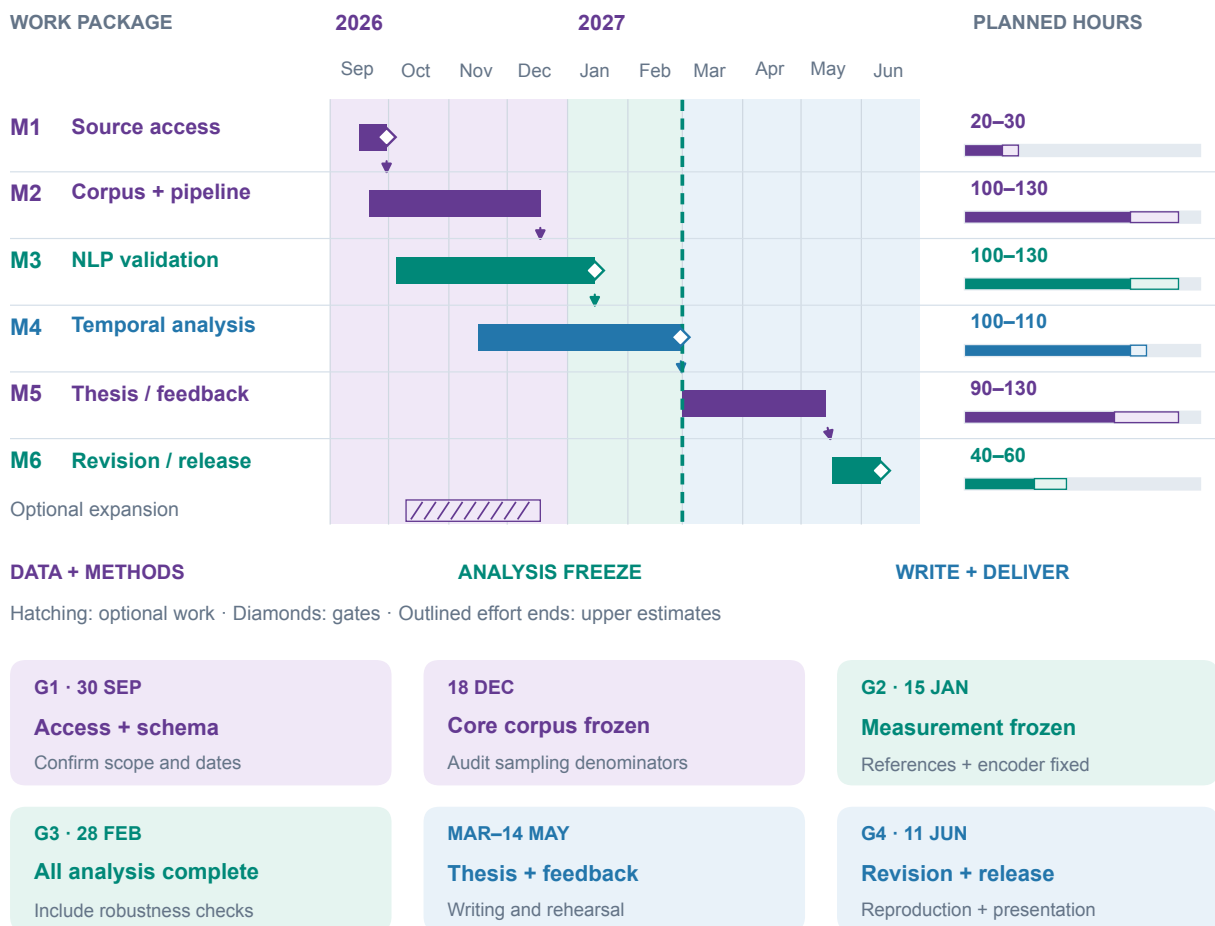


Figure 9. Work programme and effort. Solid/outlined bars show lower/upper estimates; hatching denotes optional work. Hours are planned.

Resource dependencies are permitted archives, storage/local compute, annotation time and supervisory feedback. The pilot is due by 16 October; core data and denominator checks freeze on 18 December. Measurement validation freezes on 15 January, followed by all temporal, event and structural analyses by 28 February. Release requires a permissions review and reproduction instructions.

Assessment anchors: seminar, 12-16 October; Thesis Plan, 1-19 March 2027; rehearsal, 10-14 May; final pitch/poster/demonstration/Q&A, 7-11 June. March-June centres on writing and feedback using completed analyses.

Missed freezes trigger narrower comparisons while preserving construct validation, a bounded cross-role comparison and the February target.

3.10 OHS, ethics and data governance

All proposed human-language sources and derived databases will undergo documented ethics screening before substantive use. This project commitment covers public policy/news, public posts, quotations, identifiers, embeddings, indices and release plans; public availability is not blanket permission.

Table 8. Human-data and OHS controls.

Issue	Project control	Residual check
Identifiability	Minimise identifiers; restrict originals and exact quotations	Searchable text may re-identify people
Derived data	Screen embeddings, indices, linkage and releases	Transformation is not automatic anonymisation
Independent review	Neutral guide; separate initial judgements	Reader needs permitted data access
Group representation	Observable roles; unknowns retained	No inferred sensitive identities
Workload / wellbeing	Bounded reading sessions, breaks and ergonomic setup	Monitor fatigue/distressing content

Institutional pathway. UQ requires research involving or about humans or their data to undergo ethics review or qualify for exemption. Determine the route with Research Ethics and Integrity; submit review or applicable exemption registration through MyResearch. Begin relevant work only when approval or exemption conditions are established. Because publication is intended, do not rely on a training-only exemption. Not every public institutional document necessarily requires HREC review; the applicable pathway must be documented ([The University of Queensland, n.d.](#)). No clearance is claimed here.

Independent methodological review. Use a reader unfamiliar with the project and uninvolved in selecting references or models. They receive enough neutral instruction to assess passages, but not the preferred result or initial labels. This reduces investigator expectancy; the supervisor's oversight does not substitute for independent validation. Nor can either reader or supervisor replace institutional ethics determination. Reddit source access, including access for an independent reader, requires its own permission ([Reddit, n.d.](#)).

Later human evaluation. If reviewer behaviour, A/B responses or other participant data become research outcomes, seek UQ advice on the participant role and any approval/amendment before recruitment. Document consent and permitted processing. Separate ethical clearance from copyright, platform terms and storage/export permissions; none substitutes for the others.

3.11 Outputs and research record

Release only permitted data/code with provenance, role/time keys, units, missingness, reference/model versions, edge definitions and uncertainty. Restricted originals remain separate. A dependency manifest and rollback log support reproduction; the thesis links temporal/structural findings to representative and contradictory passages. Independent review and supervisory feedback are recorded separately.

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